A logarithmic-derivative comparison for Dirichlet Newman families

Drafted for Edward Baker

17 October 2026

Research draft. Internal LLM checks include a review using another GPT model.
Independent specialist review and a priority audit remain outstanding.

Abstract

We give an explicit lower bound for the off-zero logarithmic derivative of a primitive Dirichlet heat deformation. A scalar comparison then controls the time at which all its zeros become real. A real sine tail added to canonical-product truncations preserves the required initial lower bound on the whole real line and allows the comparison to pass to the original function without a compact-confinement hypothesis. In the normalization $s = (1 + iz)/2$, the argument gives $\Lambda_\chi \le 27/100$ for every primitive character of conductor $q \ge 10^{12}$, and $\Lambda_\chi = O((\log q)^{-4/7})$ uniformly as $q \to \infty$. The finite scalar inequalities use exact rational enclosures. The analytic proof details are included in the appendices; the exact scalar programme and its output form the computational supplement. A sibling investigation obtains the stronger finite bound 43/200 using a uniform positive-time floor and a restarted comparison; the present draft retains the simpler initial-floor argument and its conductor-aspect rate.

1 Normalization and statements

Let χ be primitive of conductor $q > 1$, with $\chi(-1) = (-1)^{\mathfrak{a}}$, $\mathfrak{a} \in \{0, 1\}$. Choose a unit phase α_χ with $\alpha_\chi^2 \varepsilon(\chi) = 1$, where $\varepsilon(\chi) = \tau(\chi)/(i^{\mathfrak{a}}\sqrt{q})$, and set

$$\xi_\chi(s) = \alpha_\chi (q/\pi)^{s/2} \Gamma\big((s + \mathfrak{a})/2\big) L(s, \chi), \tag{1}$$

$$\Phi_\chi(u) = 4\alpha_\chi (\pi/q)^{\mathfrak{a}/2} e^{(1+2\mathfrak{a})u} \sum_{n \ge 1} \chi(n) n^{\mathfrak{a}} e^{-\pi n^2 e^{4u}/q}, \tag{2}$$

$$H_{t,\chi}(z) = \int_{\mathbb{R}} e^{tu^2} \Phi_\chi(u) e^{izu}\, \mathrm{d}u. \tag{3}$$

Then $H_{0,\chi}(z) = \xi_\chi((1 + iz)/2)$ and $\partial_t H_{t,\chi} = -\partial_z^2 H_{t,\chi}$. The Hermitian functional equation makes $H_{t,\chi}$ real entire, although it need not be even. With compatible phases, $H_{t,\bar\chi}(z) = \overline{H_{t,\chi}(-\bar z)}$. The normalization is detailed in Note 1 [5] and agrees with Dobner's framework [1]. Its associated constant Λ_χ is characterized by reality of all zeros exactly for $t \ge \Lambda_\chi$.

Theorem 1.1 (Dirichlet consequence)**.** *Every primitive Dirichlet character of conductor $q \ge 10^{12}$ satisfies*

$$\Lambda_\chi \le \frac{27}{100}.$$

Moreover, uniformly over primitive characters as $q \to \infty$,

$$0 \le \Lambda_\chi = O\big((\log q)^{-4/7}\big), \qquad \Lambda_\chi \longrightarrow 0.$$

These statements use neither GRH, a quasi-RH half-plane, nor an assumed confinement theorem for the character flow.

The lower bound is Dobner's published theorem [1]. The upper bound will follow from a general comparison. For $C > 0$ define

$$D_C(t) = 1 + 2C(1 - e^{-t/2}), \qquad m_C(t) = \frac{Ce^{-t/2}}{D_C(t)}, \tag{4}$$

$$I_C(t) = e^{-t} D_C(t)^{4/3}, \qquad J_C(t) = \int_0^t I_C(s)\,\mathrm{d}s. \tag{5}$$

Theorem 1.2 (Comparison without confinement)**.** *Let F be a nonzero real entire function of order at most one, with all zeros in $|\operatorname{Im} z| \le 1$. Suppose*

$$-\operatorname{Im}\frac{F'}{F}(x + 3i) \ge C > 0 \qquad (x \in \mathbb{R}).$$

If $0 < T \le 1/2$ and $J_C(T) \ge 1/2$, then $e^{-T\partial_z^2} F$ has only real zeros.

The proof first treats flows with confined nonreal zeros, then removes that restriction by a sine-tail approximation. No evenness assumption is present. Appendices A–C supply the analytic details; Notes 2–3 [6, 7] retain their research history.

2 Canonical sums and a positive-time lower bound

For a real entire function with its zeros $\rho = a + iv$ in a bounded strip, the genus-one product gives, at p above that strip,

$$S(x, p, t) := -\operatorname{Im}\frac{F_t'}{F_t}(x + ip) = \sum_\rho \frac{p - v}{(x - a)^2 + (p - v)^2}. \tag{6}$$

Zeros are counted with multiplicity, including a possible zero at the origin. The imaginary compensators cancel under conjugation. Their absolute summability follows from bounded imaginary parts and $\sum_{\rho \ne 0} |\rho|^{-2} < \infty$. A separate, possibly divergent, real sum of reciprocal zeros is never used.

For the Dirichlet flow, the double-exponential kernel decay gives

$$\log \max_{|z| \le R} |H_{t,\chi}(z)| = O_{\chi,T}((1 + R)\log(2 + R)) \quad (0 \le t \le T). \tag{7}$$

The approximants in Section 4 have exponential-type growth. These bounds justify the canonical differentiations and completeness of the real characteristics below, as proved in Appendix A. No finite-type assumption is imposed on the original F of Theorem 1.2: this characteristic argument is applied to its fixed approximants.

Proposition 2.1 (Burgers lower bound)**.** *For the Dirichlet flow or the exponential-type approximants below, suppose the zeros remain in $|\operatorname{Im} z| \le 1$ and $S(x, p, 0) \ge C > 0$. Put $d = p - 1 > 0$. Then*

$$S(x, p, t) \ge \frac{Ce^{-2t/d^2}}{1 + dC(1 - e^{-2t/d^2})} \qquad (x \in \mathbb{R},\ 0 \le t \le 1/2). \tag{8}$$

Proof. Write $U = F_t'/F_t = u - iS$ on the line of height p. Equation (6) and its differentiated sums give

$$|u_x| \le S/d, \qquad |S_{xx}| \le 2S/d^2.$$

The heat equation implies $U_t = -U_{xx} - 2UU_x$, hence

$$S_t + 2uS_x = -S_{xx} - 2Su_x.$$

Along $x' = 2u$, this yields $S' \geq -2S/d^2 - 2S^2/d$. The Riccati solution with initial value C is the right side of (8). The characteristics are complete in both time directions: growth and a uniformly zero-free anchor ip imply $|u(x,p,t)| = O((1+|x|)\log(2+|x|))$ on bounded time intervals, which prevents finite-time escape. Their flow is onto $\mathbb{R}$, proving uniformity in x. □

At $p = 3$, the right side is $m_C(t)$ in (4). This is a quantified loss, rather than an assertion that the time-zero lower bound persists unchanged.

3 The comparison when contacts are compact

We briefly retain the confinement hypothesis used in Note 2: on each $[\delta, T]$, $\delta > 0$, the nonreal zeros lie in a bounded horizontal region. This hypothesis will be established for approximants only.

At a highest simple zero $x + ih$, put $Q = h^2$. The conjugate-pair kernel

$$K_h(d,v) = \frac{h-v}{d^2 + (h-v)^2} + \frac{h+v}{d^2 + (h+v)^2}$$

obeys $K_h/h \geq K_p/p$ whenever $|v| \leq h$, $p^2 \geq 5h^2$, and $(d,v) \neq (0, \pm h)$. Clearing denominators reduces this to nonnegativity of

$$D^2 + (A + B - 6r)D + (A-r)(B-r), \quad D = d^2,\ A = h^2,\ B = p^2,\ 0 \leq r = v^2 \leq A.$$

The zero-motion identity, with the zero's own conjugate pair removed, therefore gives

$$Q' \leq -2 - \frac{4Q}{p} S(x,p,t) + \frac{8Q}{p^2 - Q}. \tag{9}$$

This form of the comparison appears in Hakimoglu-Brown [4]; the algebra and its non-even extension are recorded in Notes 1–2.

Classical strip contraction gives $Q \leq 1$. At $p = 3$, $8/(9-Q) \leq 1$, so Proposition 2.1 gives

$$Q' \leq -2 - \left(\frac{4}{3} m_C(t) - 1\right) Q.$$

The corresponding solution from one is

$$\overline{Q}(t) = \frac{1 - 2J_C(t)}{I_C(t)}. \tag{10}$$

Until it reaches zero it decreases, with descent rate at least $2 - \overline{Q} \geq 1$.

For completeness, use a strictly higher comparison curve, with initial value $1 + \varepsilon$ and a positive ε added to its differential equation. This supplies an initial time gap uniformly in space. Confinement makes a later first contact compact. A simple contact contradicts (9); a multiple contact has a backward Hermite split above the smooth curve. Letting $\varepsilon \downarrow 0$ yields $M(t) \leq \overline{Q}(t)$ while the latter is positive. If it reaches zero at T_0, strip contraction makes $F_{t + \overline{Q}(t)/2}$ real-rooted for $t < T_0$. Heat continuity and Hurwitz then make F_{T_0} real-rooted; strip contraction preserves that property afterwards. Lemma B.1 below gives the compact-contact argument, including the multiple-zero case and the endpoint.

4 Removing confinement by a real sine tail

Fix $0 < C_1 < C$, where C is the input in Theorem 1.2, and write

$$F(z) = ce^{az}z^m \prod_{\rho\neq 0} E_1(z/\rho), \qquad E_1(w) = (1-w)e^w, \quad c, a \in \mathbb{R}.$$

Let P_R truncate this product at $|\rho| \le R$, retaining conjugate pairs and multiplicities. Set

$$\kappa = 2(C_1+1), \quad h = \pi/\kappa, \quad M = \lfloor R/(4h) \rfloor, \qquad G_R(z) = P_R(z) \prod_{k>M} \left(1 - \frac{z^2}{k^2h^2}\right). \tag{11}$$

The new zeros are real, and $G_R \to F$ locally uniformly.

Lemma 4.1. *For all sufficiently large R, $S_{G_R}(x, 3, 0) \ge C_1$ for every real x. Each fixed approximant flow has uniformly confined nonreal zeros on every bounded time interval. Moreover, $(G_R)_t \to F_t$ locally uniformly, uniformly on bounded nonnegative time intervals.*

Proof. We give the estimates here and prove the approximation and heat-limit details in Appendix C. For $|x| \le R/2$, the omitted true-zero Poisson tail is at most

$$16(p+1) \sum_{|\rho|>R} |\rho|^{-2} = o(1).$$

For $|x| \ge R/2$, the full real lattice has lower bound $\kappa \tanh(\kappa p)$, and removing its central zeros costs at most $8p/(hR) + 16p/R^2$. At $p = 3$ this leaves a bound exceeding C_1; the strict input gap $C - C_1$ handles the central region.

For fixed R, the sine product rewrites G_R as $e^{b_R z} r_R(z) \sin(\kappa z)$ with b_R real and r_R real rational, its poles canceled by sine zeros. If $r_R(z) \sim c_R z^{d_R}$ at infinity, Gaussian imaginary translation gives, uniformly for $|\operatorname{Im} z| \le 1$ and bounded t,

$$\frac{(G_R)_t(z)}{c_R z^{d_R} e^{b_R z + (\kappa^2 - b_R^2)t}} = \sin\big(\kappa(z - 2b_R t)\big) + O_{R,T}(|\operatorname{Re} z|^{-1}). \tag{12}$$

Along the Gaussian line the real part remains $\operatorname{Re} z$, so no rational pole is crossed. Rouché disks about the distant real sine zeros contain exactly one zero each; reality forces it to be real. The complement in the strip is zero-free. The strip itself follows from the genus-one operator theorem of Cardon [2, Theorem 1.5], with quadratic parameter $2t$ and no quadratic exponential in G_R.

Finally, for any fixed $1 < \sigma < 2$, canonical-product bounds give $|P_R(z)| \le e^{A(1+|z|)^\sigma}$ uniformly in R. The added tail is bounded by $\sinh(\kappa|z|)/(\kappa|z|) \le e^{\kappa|z|}$. These provide a common sub-Gaussian majorant in

$$F_t(z) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-v^2} F(z + 2i\sqrt{t}\, v)\, \mathrm{d}v.$$

Dominated convergence proves the heat-limit assertion. The limit is nonzero: real Gaussian translation inverts imaginary Gaussian translation under this same majorant, by Lemma C.1. □

Apply the compact comparison to G_R. If $J_C(T) > 1/2$, choose $C_1 < C$ close enough that $J_{C_1}(T) > 1/2$. The lemma makes $(G_R)_T$ real-rooted. Hurwitz's theorem in each open half-plane passes this to the nonzero limit F_T. For equality, first use $T + \eta < 1/2$ and then let $\eta \downarrow 0$; the case $T = 1/2$ is classical strip contraction. This proves Theorem 1.2. No bound uniform in R on the approximants' confinement regions is required.

5 Arithmetic input and exact scalar arithmetic

The initial Dirichlet logarithmic derivative obeys

$$S_\chi(x,p,0) = \frac{1}{2}\operatorname{Re}\frac{\xi_\chi'}{\xi_\chi}\left(\frac{1+p+ix}{2}\right), \tag{13}$$

$$S_\chi(x,p,0) \ge C_{q,\mathfrak{a}}(p) := \frac{1}{4}\left[\log\frac{q}{\pi} + \psi\left(\frac{1+p+2\mathfrak{a}}{4}\right)\right] - \frac{1}{2}\left(-\frac{\zeta'}{\zeta}\right)\left(\frac{1+p}{2}\right). \tag{14}$$

Indeed, the Hermitian functional equation gives (13); the gamma logarithmic derivative is bounded below by its value on the real axis, and the Euler product gives $|L'/L(s,\chi)| \le -\zeta'/\zeta(\operatorname{Re} s)$ for $\operatorname{Re} s > 1$. At $p = 3$ the even case is the smaller of the two parities.

The finite certificate in Note 2, Section 5 [6], proves

$$C_{q,\mathfrak{a}}(3) > \frac{1221}{200} = 6.105 > 6 \qquad (q \ge 10^{12},\ \mathfrak{a} = 0,1). \tag{15}$$

It uses $\log 10 > 23/10$, $\log 4 < 7/5$, $\pi < 4$, $\gamma < H_{1000} - \log 1000 < 29/50$, and a prime-power sum through 1000 with the positive tail $(\log 1000 + 1)/1000$ to prove $(-\zeta'/\zeta)(2) < 3/5$. All decisions use exact rational enclosures.

The integral in (5) has the elementary form

$$J_C(t) = \frac{3(1+2C)}{14C^2}(D_C(t)^{7/3} - 1) - \frac{3}{20C^2}(D_C(t)^{10/3} - 1). \tag{16}$$

For $C = 6$, $T = 27/100$, the same certificate encloses

$$0.502464325582617834 \le J_6(T) \le 0.502464325582617835.$$

Alternating exponential series and rational cube-root bisection make the lower endpoint rigorous. Theorem 1.2 and (15) prove the explicit part of Theorem 1.1. The computational supplement consists of `certify_dirichlet_scalar_conditional_20261017.py` and its JSON output in the investigation's `numerics/` directory. It uses Python's rational arithmetic; the deciding bounds and input parameters are those just displayed. The build record pins their hashes. The historical word "conditional" in their filenames records the original analytic contract; Section 4 supplies it without changing the arithmetic.

For $0 \le t \le 1/2$,

$$J_C(t) \ge e^{-1/2}\frac{(1 + Ce^{-1/4}t)^{7/3} - 1}{(7/3)Ce^{-1/4}}.$$

Consequently

$$B(C) := \frac{e^{1/4}}{C}\left[\left(1 + \frac{7}{6}e^{1/4}C\right)^{3/7} - 1\right] \tag{17}$$

is a sufficient reality time whenever $B(C) \le 1/2$ and $C \le C_{q,\mathfrak{a}}(3)$. Since $C_{q,\mathfrak{a}}(3) = \frac{1}{4}\log q + O(1)$ uniformly in parity, $B(C) = O((\log q)^{-4/7})$ for the natural input. This proves the remaining assertions of Theorem 1.1.

6 Scope, controls, and provenance

The sine-tail limit bypasses confinement for the original Dirichlet flow; it does not prove that separate assertion. A periodic real-entire heat flow may have a positive initial logarithmic-derivative floor and unboundedly many nonreal zeros at positive times. The example and its scope are in Note 2, Section 7. This does not contradict the present quantitative comparison.

The arithmetic gain enters through (14). Euler-factor collision controls can violate its precise Euler-product allowance. Weighted-square, reflected, and primitive-compatible controls need not have the character completion or that allowance. The general comparison remains valid when their actual input floor meets its assumptions; no character-specific lower bound is assigned to them by analogy. This is smoothing at positive time, not a proof of GRH at time zero.

A literature correction is necessary. Ki–Kim–Lee [3, Section 6, Example 1] already extend their qualitative strict bound below $1/2$ to real primitive Dirichlet characters. The initial investigation's search missed that precedent. Their example for a complex character uses a real self-dual combination, rather than the individual non-even family in (3). We make no priority claim for the quantitative results of this draft.

Stronger sibling benchmark. Project 03's round-two notes [8, 9] prove in-note $S_\chi(x, 3, t) > 6$ for all real x, $0 \le t \le 1/4$ and primitive $q \ge 10^{12}$. An intermediate-strip comparison, restarted on the original flow, then gives $\Lambda_\chi \le 43/200 = 0.215$. The scalar checks use Arb enclosures and exact rational inequalities; the analytic domination and restart are essential additional inputs. That result currently has internal same-model checks. The 0.27 bound here is a simpler consequence of the initial floor, not the repository's strongest bound at this conductor threshold.

Public provenance. A MathOverflow question [10], posted on 1 October 2026, asks about the conductor limit and outlines a stronger heuristic scale. Its missing uniform positive-time estimates are explicitly acknowledged there. It is relevant provenance, not an imported theorem or a completed priority audit.

Acknowledgment and review status. This manuscript was drafted for Edward Baker with substantial assistance from OpenAI's `gpt-6.1-sol` in Codex, recorded reasoning effort `ultra`. Same-model subagents assisted with derivation and adversarial proof checks. The exact scalar programme was also generated with LLM assistance. A subsequent mathematical review and this proof expansion used OpenAI's `gpt-6-astra`, recorded effort `max`; the review used a different GPT model from the original derivation. Neither activity constitutes independent specialist review. Notes 1–3 and the dated review-response note preserve the dependencies, corrections, source identities and calculation records. Model choices record actual usage and do not prescribe assignments for future work.

A Growth and complete characteristics

Here and below a bounded time interval means a compact interval of nonnegative times. The Hermitian theta kernel in (2) satisfies

$$|\Phi_\chi(u)| \le K_\chi e^{3|u|} \exp(-c_\chi e^{4|u|}), \qquad c_\chi > 0.$$

On the positive half-line this follows from its series, and on the negative half-line from $\Phi_\chi(-u) = \overline{\Phi_\chi(u)}$, the theta transformation in Dobner's normalization. Absorbing bounded complex-time

weights and $e^{3|u|}$ into half of the double exponential bounds the integral on $|z| \le R$ by a constant times

$$1 + \int_0^\infty e^{Ru - ce^{4u}}\,\mathrm{d}u \le 1 + \tfrac{1}{4}c^{-R/4}\Gamma(R/4) \quad (R \ge 1).$$

Stirling's bound proves (7). The same domination proves joint analyticity and differentiation under the integral. Fourier injectivity excludes the zero function. For each fixed approximant, $|G_R(z)| \le K_R e^{A_R|z|}$ after enlarging A_R to absorb its polynomial factor. Gaussian translation gives this same form of bound uniformly in time. Its heat flow is nonzero by Lemma C.1.

Lemma A.1 (Complete characteristics)**.** *Suppose a nonzero real-entire heat flow F_t has all zeros in $|\operatorname{Im} z| \le 1$ and $\log \max_{|z|\le R} |F_t(z)| \le K(1+R)\log(2+R)$ uniformly on a bounded time interval. Then for $p > 1$ its real characteristic equation $x' = 2\operatorname{Re}(F_t'/F_t)(x+ip)$ has unique complete trajectories, forwards and backwards on that interval. The map between any two times is onto $\mathbb{R}$.*

Proof. The zero-free point ip is a uniform anchor: $\inf_t |F_t(ip)| > 0$ by continuity. Jensen's formula on disks centred there, together with the growth bound, gives uniform zero counts $N_t(R) = O(R\log(2+R))$. The logarithmic derivative at the anchor is uniformly bounded. Retaining canonical summands until subtracting gives the absolutely convergent identity

$$U_t(x+ip) - U_t(ip) = \sum_\rho \left(\frac{1}{x+ip-\rho} - \frac{1}{ip-\rho} \right), \qquad U_t = F_t'/F_t.$$

The possible origin zero is included. Put $d = p-1$ and choose $R \ge 2(1+|x|+p)$. The terms $|\rho| \le 2R$ contribute $O(N_t(2R)/d)$. The others contribute $O(R\sum_{|\rho|>2R} |\rho|^{-2}) = O(\log(2+R))$, by the zero count and partial summation. Consequently

$$|\operatorname{Re} U_t(x+ip)| \le K_p(1+|x|)\log(2+|x|).$$

The vector field is smooth, so local uniqueness holds. Since $\int^\infty [(1+r)\log(2+r)]^{-1}\,\mathrm{d}r = \infty$, this bound precludes escape to infinity in finite time in either direction. Solving backwards from every final point proves surjectivity. □

The canonical derivative sums converge locally uniformly off the strip. In particular $U_x = -\sum_\rho (z-\rho)^{-2}$ and $U_{xx} = 2\sum_\rho (z-\rho)^{-3}$. At $z = x+ip$, each $y = p - \operatorname{Im}\rho \ge d$ and $|z-\rho| \ge y$, so

$$|U_x| \le \sum_\rho |z-\rho|^{-2} \le S/d, \qquad |\operatorname{Im} U_{xx}| \le 2\sum_\rho |z-\rho|^{-3} \le 2S/d^2.$$

These are the estimates used in Proposition 2.1. The preceding proof applies to the Dirichlet flow and to each fixed approximant, with constants allowed to depend on that flow. It does not assert the displayed growth bound for every entire function of order one.

B The compact-contact lemma

Lemma B.1. *Let F_t be a jointly analytic, nonzero real-entire heat flow with the uniform growth bound of Lemma A.1, initial strip $|\operatorname{Im} z| \le 1$, classical strip contraction, and $S(x,3,t) \ge m_C(t)$ for all real x. Suppose the nonreal zeros are uniformly confined in the real direction on every $[\delta, T]$ with $\delta > 0$. Then, while* (10) *is positive, every zero satisfies $(\operatorname{Im} z)^2 \le \overline{Q}(t)$. If $J_C(T) \ge 1/2$, all zeros at time T are real.*

Proof. Set $a(t) = 4m_C(t)/3 - 1$. For $\epsilon > 0$, let

$$q_\epsilon' = -2 - a(t)q_\epsilon + \epsilon, \qquad q_\epsilon(0) = 1 + \epsilon.$$

On any compact interval before the first zero of $\overline{Q}$, $q_\epsilon > \overline{Q} > 0$ and $q_\epsilon \to \overline{Q}$ uniformly. Initially $q_\epsilon > 1$ on a time interval independent of position, whereas the actual squared heights never exceed one. If a zero first reaches q_ϵ, that contact therefore has positive time. Confinement and the strip put the possible contacts in a compact set. Continuity of zeros, or the argument principle on small circles, then gives an attained first contact.

For a simple contact at $z = x+ih$, $h > 0$, the implicit function theorem and the heat equation give $z' = F_t''(z)/F_t'(z)$. In the canonical product, the conjugate zero contributes $-1/h$ to h'. Summing over the remaining zeros in the closed upper half-plane, with weight one for a nonreal zero and half for a real zero, gives

$$h' = -\frac{1}{h} - 2\sum{}' \varpi_k K_h(x - \operatorname{Re}\rho_k, \operatorname{Im}\rho_k).$$

The real exponential drift contributes no imaginary part. The kernel inequality of Section 3 gives (9). At first contact the zero is highest, $h^2 = q_\epsilon \le 1$, and hence $(h^2)' \le -2 - a(t)q_\epsilon < q_\epsilon'$, a contradiction.

For a contact of multiplicity $m \ge 2$ at (t_0, z_0), write $F_{t_0}(z_0 + w) = cw^m + O(w^{m+1})$, $c \ne 0$. Joint analyticity and the heat equation imply, locally uniformly in ζ,

$$v^{-m/2} F_{t_0 - v}(z_0 + \sqrt{v}\,\zeta) \longrightarrow c\, e^{\partial_\zeta^2}\zeta^m = c \sum_{j=0}^{\lfloor m/2 \rfloor} \frac{m!\,\zeta^{m-2j}}{j!(m-2j)!}.$$

The last polynomial is a rescaled Hermite polynomial: its roots are simple and purely imaginary, and for $m \ge 2$ one has positive imaginary part. Rouché's theorem on a small circle about that root therefore gives an earlier zero with height $\operatorname{Im} z_0 + c_m\sqrt{v} + o(\sqrt{v})$, $c_m > 0$. The positive comparison height $\sqrt{q_\epsilon(t)}$ changes by $O(v)$ near t_0. This contradicts the absence of earlier contact. Thus no contact is possible, and letting $\epsilon \downarrow 0$ proves the strip bound.

Let T_0 be the first zero of $\overline{Q}$. For every $t < T_0$, strip contraction makes $F_{t+\overline{Q}(t)/2}$ real-rooted. Those times tend to T_0. Heat continuity and Hurwitz in each open half-plane make the nonzero F_{T_0} real-rooted, and strip contraction preserves reality for all later times. Since $I_C > 0$ and J_C is strictly increasing, $J_C(T) \ge 1/2$ implies $T_0 \le T$. □

C Sine tails and passage through the heat operator

C.1 The floor on the whole real line

For R large enough, $|x| \le R/2$ and $|\rho| > R$, the initial strip gives $|x - \operatorname{Re}\rho| \ge |\rho|/4$. Thus the omitted Poisson mass is at most $16(p+1)\sum_{|\rho|>R}|\rho|^{-2} = o(1)$, uniformly on that central region. All added zeros are real and contribute nonnegative mass. The strict gap $C - C_1$ gives the required central floor.

The full lattice $h\mathbb{Z}$ has mass

$$\sum_{k\in\mathbb{Z}} \frac{p}{(x-kh)^2 + p^2} = \frac{\kappa\sinh(2\kappa p)}{\cosh(2\kappa p) - \cos(2\kappa x)} \ge \kappa\tanh(\kappa p).$$

For $|x| \ge R/2$ and $|k| \le M$, $|x - kh| \ge R/4$. Removing these $2M + 1 \le R/(2h) + 1$ terms costs at most $8p/(hR) + 16p/R^2$. At $p = 3$, $\kappa\tanh(3\kappa) > C_1$ because $\kappa = 2(C_1 + 1) > 2$ and $\tanh 6 > 1/2$. The remaining true zeros again contribute nonnegative mass. This proves the global floor for large R.

C.2 Confinement for a fixed approximant

Euler's product for sine turns (11) into $G_R(z) = e^{b_R z} r_R(z) \sin(\kappa z)$, where b_R is real and r_R is real rational with only real poles, canceled by sine zeros. Write $r_R(z) \sim c_R z^{d_R}$, where d_R is an integer, possibly negative. For $z = x + iy$, $|y| \le 1$, $0 \le t \le T$, and $w = z + 2i\sqrt{t}\, v$, rational asymptotics give, for sufficiently large $|x|$,

$$\frac{r_R(w)}{c_R z^{d_R}} = 1 + O_{R,T}\left(\frac{(1+|v|)^{K_R}}{|x|}\right).$$

Indeed $\operatorname{Re} w = x$ avoids every pole, $|w| \ge |x|$, and powers of $w/z = 1 + 2i\sqrt{t}\, v/z$ have this bound for positive or negative integer exponents. The sine factor grows at most like $e^{\kappa(1+2\sqrt{T}|v|)}$ along this line, while the extra $e^{2ib_R\sqrt{t}\, v}$ has modulus one. Integrating against e^{-v^2} therefore gives a uniform $O_{R,T}(|x|^{-1})$ error. Evaluating the two elementary exponential Gaussian integrals in the leading term proves (12).

Cardon's theorem, applied to the genus-one entire G_R with initial strip one and operator $e^{-t\partial_z^2}$, confines all zeros to $|\operatorname{Im} z| \le \sqrt{\max(1-2t, 0)}$. Choose a fixed disk radius $0 < r < \min(h/4, 1/2)$ about each distant sine zero $kh + 2b_R t$. On each boundary the leading sine has a positive lower modulus independent of k and t. Equation (12) and Rouché give exactly one zero in every sufficiently distant disk. Conjugation fixes each disk and the function is real entire, so that single zero is real. On the remaining part of $|\operatorname{Im} z| \le 1$ the sine is bounded away from zero, so the same estimate excludes other distant zeros. The required cutoff may depend on R and T. This proves uniform confinement for each fixed approximant, even down to time zero.

C.3 A common majorant, nonvanishing and the limit

Lemma C.1. *The approximants converge under Gaussian heat translation locally uniformly in z and uniformly on bounded nonnegative time intervals. Every heat translate of the original F and of an approximant is nonzero. The heat translations form a semigroup on these functions.*

Proof. Fix $1 < \sigma < 2$. An order-at-most-one zero set has $\sum_{\rho \ne 0} |\rho|^{-\sigma} < \infty$. The elementary bound $|E_1(w)| \le \exp(C_\sigma |w|^\sigma)$ follows from its quadratic Taylor estimate for $|w| \le 1$ and $|1 - w|e^{|w|}$ for $|w| > 1$. Consequently, including $ce^{az}z^m$, $|P_R(z)| \le \exp(A(1+|z|)^\sigma)$ with A independent of R. For the tail,

$$\left|\prod_{k>M}\left(1 - \frac{z^2}{k^2h^2}\right)\right| \le \prod_{k\ge 1}\left(1 + \frac{|z|^2}{k^2h^2}\right) = \frac{\sinh(\kappa|z|)}{\kappa|z|} \le e^{\kappa|z|}.$$

The omitted lattice tail tends to one locally uniformly, because $\sum_{k>M} k^{-2} \to 0$. Hence $G_R \to F$ locally uniformly and all have a common sub-Gaussian majorant. In

$$(G_R)_t(z) = \pi^{-1/2} \int_{\mathbb{R}} e^{-v^2} G_R(z + 2i\sqrt{t}\, v)\, \mathrm{d}v$$

this majorant is integrable uniformly for z in a compact set and $0 \le t \le T$. Splitting into bounded v and the uniform tail proves the asserted convergence, including $t = 0$.

For any such input F, Gaussian translation also preserves a bound of order $\sigma < 2$. Real translation inverts its imaginary heat translation: absolute convergence permits Fubini in

$$\pi^{-1} \int_{\mathbb{R}^2} e^{-u^2-v^2} F(z + 2\sqrt{t}\, u + 2i\sqrt{t}\, v)\, \mathrm{d}u\, \mathrm{d}v = F(z).$$

In polar coordinates the angular mean is $F(z)$ by the mean-value identity for an entire function, and the radial integral has mass one. Thus $F_t \equiv 0$ would imply $F \equiv 0$. Finally two imaginary

Gaussian translations combine by an orthogonal change of the two real Gaussian variables, giving $(F_t)_s = F_{t+s}$. All exchanges are justified by the same sub-Gaussian domination. □

These lemmas close the analytic proof of Theorem 1.2. Its finite arithmetic remains reproducible from the identified computational supplement; the local Markdown notes are historical references rather than prerequisites for the analytic argument.

References

[1] A. Dobner, *A proof of Newman's conjecture for the extended Selberg class*, Acta Arithmetica **201** (2021), 29–62; arXiv:2005.05142v2.

[2] D. A. Cardon, *Complex zero strip decreasing operators*, arXiv:1401.0237v2, Theorem 1.5; J. Math. Anal. Appl. (2015), doi:10.1016/j.jmaa.2015.01.026.

[3] H. Ki, Y.-O. Kim, and J. Lee, *On the de Bruijn–Newman constant*, Advances in Mathematics **222** (2009), 281–306, doi:10.1016/j.aim.2009.04.003.

[4] C. Hakimoglu-Brown, *A conditional bound* $\Lambda \leq 0.13963$ *for the de Bruijn–Newman constant*, arXiv:2610.12803v1 (2026), Section 4.

[5] Research Note 1, prepared for E. Baker with Claude Opus 5.5, *Normalization and transfer contract*, 17 October 2026, notes/round-1.

[6] Research Note 2, prepared for E. Baker with GPT-6.1-sol, *An explicit positive-time floor and a comparison conditional on confinement*, 17 October 2026, notes/round-2.

[7] Research Note 3, prepared for E. Baker with GPT-6.1-sol, *Removing confinement from the strip comparison by a real sine tail*, 17 October 2026, notes/round-3.

[8] Research Note 1, Project 03, round 2, prepared for E. Baker with GPT-6-astra, *A uniform height three floor for large conductor characters*, 17 October 2026, Project 03, uniform-floor note.

[9] Research Note 2, Project 03, round 2, prepared for E. Baker with GPT-6-astra, *A restarted strip comparison consuming a global positive-time floor*, 17 October 2026, Project 03, restarted-comparison note.

[10] MathOverflow user khonsu, *Do de Bruijn–Newman constants of Dirichlet L-functions tend to 0 as the conductor tends to infinity?*, question 515667, posted 1 October 2026, edited 2 October 2026; MathOverflow question.